

The Art of Computer Programming

ERRATA ET ADDENDA

October 19 1979

This list supplements previous errata published in Stanford reports CS551 (1976) and CS712 (1979). If you want to print your own copy using the SU-AI computer, do "XS ERRATA.XGP [ART,DEK] ".

1.1 historical improvements 9/4/79 1

lines -5, -4: Khowârizm ... Khiva. $\rightsquigarrow$ The Aral Sea in Central Asia was once known as Lake Khowârizm, and the Khowârizm region is located in the Amu River basin just south of that sea.

line -3: *muqabala* $\rightsquigarrow$ *muqâbala*

line -3: restoration and reduction $\rightsquigarrow$ restoring and equating

lines -2, -1: although ... algebraic. $\rightsquigarrow$ which was a systematic study of the solution of linear and quadratic equations.

1.88 bottom line 4/1/79 2

$1 \leq j < m$ $\rightsquigarrow$ $0 \leq j < m$

1.246 improved overlap 2/4/79 3

line -10 should become: $\text{OLDTOP}[j] \equiv D[j] \equiv \text{NEWBASE}[j + 1]$

line -9: $n + 1$; $\rightsquigarrow$ n ;

lines -8 and -7: delete the sentence "It will ... overlap."

1.248 addendum to 1979 change #47 2/7/79 4

See also A. S. Fraenkel, *Inf. Proc. Letters* 8 (1979), 9–10, who suggests working with pairs of stacks that grow towards each other.

1.250
new rating for exercise 13
3/1/79
5

[M47]
↗
[M49]

1.254
new lines 16 and 17
2/4/79
6

Otherwise set $X \leftarrow \text{POOLMAX}$ and $\text{POOLMAX} \leftarrow \text{POOLMAX} + c$, (7)
where c is the node size; **OVERFLOW** now occurs if $\text{POOLMAX} > \text{SEQMIN}$.”

1.284
the line for time 0693
7/1/79
7

M1
↗
M5

1.323
trivial improvements to Program S
10/17/79
8

line 03: ENT6
↗
ENT5
line 03: Q
↗
P
line 04: S2
↗
2F
line 09: $n + 1$
↗
 n
line 09: Set
↗
S2 Search to left Set
line 10, first column:
↗
2H
line 11: *-2
↗
S2

1.324
line 5
10/17/79
9

8
↗
7

1.405
new exercise
10/18/79
10

13. [M25] (F. K. Hwang.) Let $w_1 \leq \cdots \leq w_m$ and $w'_1 \leq \cdots \leq w'_m$ be two sets of weights with $\sum_{1 \leq j \leq k} w_j \leq \sum_{1 \leq j \leq k} w'_j$ for $1 \leq k \leq m$. Prove that the costs of the corresponding minimum weighted path lengths satisfy $\sum_{1 \leq j \leq m} w_j l_j \leq \sum_{1 \leq j \leq m} w'_j l_j$.

1.420
new paragraph before the exercises
2/7/79
11

Daniel P. Friedman and David S. Wise have observed that the reference counter method can be employed satisfactorily in many cases even when lists point to themselves, if certain link fields are not included in the counts [*Inf. Proc. Letters* 8 (1979), 41–45].

1.463 **correction to 1979 change #73** 2/14/79 12

Such graph machines ... fixed. $\swarrow$ Linking automata can easily simulate graph machines, taking at most a bounded number of steps per graph step. Conversely, however, it is unlikely that graph machines can simulate arbitrary linking automata without unboundedly increasing the running time, unless the definition is changed from undirected to directed graphs, in view of the restriction to vertices of bounded degree.

1.515 **line 5** 10/18/79 13

For ... history, $\swarrow$

Historical notes: C. Haros gave a (more complicated) rule for constructing such sequences, in *J. de l'École Polytechnique* 4, 11 (1802), 364–368; his method was correct, but his proof was inadequate. The geologist John Farey independently conjectured several years later that x_k/y_k is always equal to $(x_{k-1} + x_{k+1})/(y_{k-1} + y_{k+1})$ [*Philos. Magazine and Journal* 47 (1816), 385–386]; a proof was supplied shortly afterwards by A. Cauchy [*Bull. Société Philomathique de Paris* (3) 3 (1816), 133–135], who attached Farey's name to the series. For more of its interesting properties,

1.531 **line -2** 10/18/79 14

X's. For the history of the ballot problem $\swarrow$ X's. This problem was actually resolved as early as 1708 by Abraham de Moivre, who showed that the number of sequences containing l A's and m B's, and containing at least one initial substring with n more A's than B's, is $f(l, m, n) = \binom{l+m}{\min(m, l-n)}$. In particular, $a_n = \binom{2n}{n} - f(n, n, 1)$ as above. (De Moivre stated this result without proof [*Philos. Trans.* 27 (1711), 262–263]; but it is clear from other passages in his paper that he knew how to prove it, since the formula is obviously true when $l \geq m + n$, and since his generating-function approach to similar problems yields the symmetry condition $f(l, m, n) = f(m + n, l - n, n)$ by simple algebra.) For the later history of the ballot problem

1.538 **insert new answer** 3/1/79 15

13. A. C. Yao has shown that $\max(k_1, k_2)$ will be $\frac{1}{2}m + (2\pi(1 - 2p))^{-1/2}\sqrt{m} + O(m^{-1/2}\log m)$ for large m , when $0 < p < \frac{1}{2}$. (To appear.)

1.548 **first line of answer 9** 4/17/79 16

should. $\swarrow$ should; except in the instructive anomalous case that $\text{COEF} = 0$ for some term with $\text{ABC} \geq 0$, when it fails badly.

1.560 additional sentence to follow 1976 change #135 1/17/79 17

(Steps T4 and T5 can be streamlined so that nodes are not taken off the stack and immediately reinserted.)

1.562 answer 21 10/17/79 18

21. The following $\swarrow \rightarrow$

21. (Solution by Ď. Branislav, traverses either in preorder or inorder.)

U1. [Initialize.] If $T = \Lambda$, terminate the algorithm. Otherwise set $Q \leftarrow T$.

U2. [Preorder visit.] If traversing in preorder, visit $\text{NODE}(Q)$.

U3. [Go to left.] Set $R \leftarrow \text{LLINK}(Q)$. If $R = \Lambda$, go to U5.

U4. [Insert a right thread.] Set $P \leftarrow Q$ and $Q \leftarrow R$, then set $R \leftarrow \text{RLINK}(R)$ zero or more times until $\text{RLINK}(R) = \Lambda$. Set $\text{RTAG}(R) \leftarrow \text{“+”}$ and $\text{RLINK}(R) \leftarrow P$. Return to step U2.

U5. [Inorder visit.] If tranersing in inorder, visit $\text{NODE}(Q)$.

U6. [Go to right.] If $\text{RLINK}(Q) \neq \Lambda$ and $\text{RTAG}(Q) = \text{“+”}$, set $Q \leftarrow \text{RLINK}(Q)$ and go to step U2.

U7. [Remove the thread.] Set $R \leftarrow \text{RLINK}(Q)$, $\text{RTAG}(Q) \leftarrow \text{“+”}$, $\text{RLINK}(Q) \leftarrow \Lambda$.

U8. [Go up.] Set $Q \leftarrow R$. Go back to U5 if $Q \neq \Lambda$, otherwise terminate the algorithm.

■

Alternatively, the following slightly slower

1.562 amendments to Algorithm V 10/17/79 19

steps V1 and V7: $\text{LOC}(T) \swarrow \rightarrow \Lambda$

step V3: delete “(It is)”

1.562
new answer 22 (extends to page 563)
10/17/79
20

22. Let $rI4 \equiv R$, $rI5 \equiv Q$, $rI6 \equiv -P$; use other conventions of Programs T and S.

01	U1	LD5	T	1	<u>U1 Initialize</u> $Q \leftarrow T$.
02		J5NZ	U3	1	
03		JMP	DONE	0	Special exit if $T = 0$
04	U4	ENN6	0,5	$a-1$	<u>U4 Insert a right thread</u> $P \leftarrow Q$.
05		ENT5	0,4	$a-1$	$Q \leftarrow R$.
06	4H	ENT3	0,4	$n-b$	$S \leftarrow R$.
07		LD4	1,3(RLINK)	$n-b$	$R \leftarrow \text{RLINK}(S)$.
08		J4NZ	4B	$n-b$	Repeat until $R = \Lambda$.
09		ST6	1,3(RLINKT)	$a-1$	$\text{RLINKT}(S) \leftarrow -P$.
10	U3	LD4	0,5(LLINK)	n	<u>U3 Go to left</u> $R \leftarrow \text{LLINK}(Q)$.
11		J4NZ	U4	n	To U4 if $R \neq \Lambda$.
12	U5	JMP	VISIT	n	<u>U5 Inorder visit</u>
13	U6	ENT4	0,5	n	<u>U6 Go to right</u> $R \leftarrow Q$.
14		LD5	1,5(RLINKT)	n	$Q \leftarrow \text{RLINKT}(Q)$.
15		J5P	U3	n	To U3 if $Q > 0$.
16	U7	STZ	1,5(RLINKT)	a	<u>U7 Remove the thread</u>
17	U8	ENN5	0,5	a	<u>U8 Go up</u> $Q \leftarrow -Q$.
18		J5NZ	U5	a	To U5 if $Q \neq \Lambda$. ■

Note that the search in step U4 is not time-consuming, since it examines each **RLINK** at most once. The total running time is $12n + 8a - 4b - 2$, where $n > 0$ is the number of nodes, a is the number of null **RLINK** s, and b is the number of nodes on the tree's "right path" T , $\text{RLINK}(T)$, $\text{RLINK}(\text{RLINK}(T))$, etc. Thus, the algorithm is competitive with that of exercise 20. The running time of an analogous program based on Algorithm V of exercise 21 is $22n - 10$.

1.579
in the biggest matrix
5/1/79
21

change the label on row 3 and the label on column 3 from [10] to [20]

1.579
in the second-biggest matrix, row 1
5/1/79
22

a_{0m} $\nwarrow$ a_{0n}

1.582
line -9
3/1/79
23

Note: Kruskal's $\nwarrow$ Note: Kruskal actually proved a stronger result, using a weaker form of embedding. His

1.589 **line 1 of answer 4** 10/18/79 24
 $l_j > l_{j+1}$  $l_j \geq l_{j+1}$

1.590 **addendum to answer 10** 10/18/79 25

(place the figure at the right margin and set the copy narrower, to its left)

The desired ternary tree is 

The desired ternary tree is shown at the right.

F. K. Hwang has observed [*SIAM J. Appl. Math.* **37** (1979), 124–127] that a similar procedure is valid for minimum weighted path length trees having any prescribed multiset of degrees: at each step the smallest t weights are combined, where t is as small as possible.

1.590 **new answer** 10/18/79 26

13. Let $l_{m+1} = l'_{m+1} = 0$. Then

$$\begin{aligned} \sum_{1 \leq j \leq m} w_j l_j &\leq \sum_{1 \leq j \leq m} w_j l'_j = \sum_{1 \leq k \leq m} (l'_j - l'_{j+1}) \sum_{1 \leq j \leq k} w_j \\ &\leq \sum_{1 \leq k \leq m} (l'_j + l'_{j+1}) \sum_{1 \leq j \leq k} w'_j = \sum_{1 \leq j \leq m} w'_j l'_j, \end{aligned}$$

since $l'_j \geq l'_{j+1}$ as in exercise 4. The same proof holds for many other kinds of optimum trees, including those of exercise 10.

1.603 **new version of lines 22–24 (overrides previous changes)** 10/18/79 27

[This method is called the “LISP 2 garbage collector.” An interesting alternative, which does not require the LINK field at the beginning of a node, can be based on the idea of linking together all pointers that point to each node—see Lars-Erik Thorelli, *BIT* **16** (1976), 426–441; F. Lockwood Morris, *CACM* **21** (1978), 662–665; and H. B. M. Jonkers, *Inf. Proc. Letters* **9** (1979), 26–30. Other methods have been published by B. K. Haddon and W. M. Waite, *Comp. J.* **10** (1967), 162–165; B. Wegbreit, *Comp. J.* **15** (1972), 204–208; D. A. Zave, *Inf. Proc. Letters* **3** (1975), 167–169.]

1.618R **entry for Binary trees, copying of** 10/17/79 28
332  331–332

1.619L		10/17/79	29
Branislav, Ďurian, 562.			
1.619R	Cauchy entry	10/18/79	30
add p. 515			
1.620R	entry for Copy a ... tree	10/17/79	31
332		331–332	
1.622R	Farey entry	10/18/79	32
add p. 515			
1.623L		2/7/79	33
Fraenkel, Aviezri S., 248.			
1.623L		2/7/79	34
Friedman, Daniel Paul, 420.			
1.623R		10/18/79	35
Haros, Ch., 515.			
1.624L		10/18/79	36
Hwang, Frank Kwangming, 405, 590.			
1.625L		10/18/79	37
Jonkers, Henricus Bernardus Maria, 603.			
1.627R		10/18/79	38
Morris, F. Lockwood, 603.			
1.633L	entry for Trees, copying of	10/17/79	39
332		331–332	
1.634L	Wise entry	2/7/79	40
add p. 420			

1.634R		3/1/79	41
Yao, Andrew Chi-Chih, 538.			
3.9	exercise 17	1/31/79	42
How	$\rightsquigarrow$ (This n is called the <i>index</i> of b modulo p , with respect to a .)	How	
3.23	lines 17 and 22	10/18/79	43
<i>Anuyogadvarā</i> $\rightsquigarrow$ <i>Anuyogadvāra</i> (twice)			
3.76	line -7	10/18/79	44
$p(n)$ $\rightsquigarrow$ $p(N)$			
3.90	caption	10/18/79	45
Fig. 12 $\rightsquigarrow$ Fig. 12.			
3.108	line -14	2/7/79	46
between $\rightsquigarrow$ between			
3.321	line 4	10/5/79	47
individual $\rightsquigarrow$ individually			
3.422	exercise 27 line 6	1/24/79	48
n $\rightsquigarrow$ k			
3.472	line 11	1/31/79	49
$\log_2$ $\rightsquigarrow$ $\lg$			
3.476	clarifications	1/31/79	50
line -14: new node $\rightsquigarrow$ new key			
line -11: nodes $\rightsquigarrow$ internal nodes			
line -10: nodes $\rightsquigarrow$ internal nodes			
line -8: a node $\rightsquigarrow$ a node while building a tree of N keys			

3.480 exercise 5 2/23/79 51

flowing.”) ↗ flowing”; pass up the key that makes the remaining two parts most nearly equal in size.)

3.491 Figure 33 2/23/79 52

(it would be desirable to show the 5-bit binary codes in fine print under the TEXT line; to make room, “TEXT: ” should be brought up to a line by itself)

3.491 line -12 2/23/79 53

contains the number 24 (the ↗ would contain the number 24 (which indicates the

3.491 line -10 2/23/79 54

$\log_2$ ↗ $\lg$

3.492 clarifications 2/23/79 55

line 4: root node, ↗ root node α ,
line 5: next node tells ↗ next node γ tells
line 7: next node tells ↗ next node ϵ tells
line 8: is 0; now we find $\text{LTAG} = 1$, ↗ is 0, so again we want to move left;
 but now we find that the pointer is a dotted line (i.e., $\text{LTAG} = 1$),
line 11: which ... pattern. ↗ having that pattern in our data.

3.506 line 8 1/24/79 56

Section ↗ Sections

3.507 update to 1979 change #259 3/1/79 57

850 ↗ 850, 22 (1979), 104,

3.543 new rating for exercise 10 3/1/79 58

$[M43]$ ↗ $[M33]$

3.544
exercise 14 (replacement for lines 3 and following)
2/23/79
59

2-bit TAG field and two link fields called LINK and AUX, with the following interpretation:

- TAG(P) = 0 indicates a word in the list of available space; LINK(P) points to the next entry in this list, and AUX(P) is unused.
- TAG(P) = 1 indicates any word in use where P is not the hash address of any key in the scatter table; the other fields of the word in location P may have any desired format.
- TAG(P) = 2 indicates that P is the hash address of at least one key; AUX(P) points to a linked list specifying all such keys, and LINK(P) points to another word in the list memory. Whenever a word with TAG(P) = 2 is accessed during the processing of any list, it is necessary to set $P \leftarrow \text{LINK}(P)$ repeatedly until reaching a word with $\text{TAG}(P) \leq 1$. (For efficiency we might also then change prior links so that it will not be necessary to skip over the same scatter table entries again and again.)

Show how to define suitable algorithms for inserting and retrieving keys in a combined table of this sort.

3.544
exercise 23
2/23/79
60

[23]

↗

[33]

3.576
addendum to 1976 change #324
1/31/79
61

use change #401 in the 1979 list

3.608
line -8
2/15/79
62

$z^{N+1-\delta_{s1}}$

↗

z^{N+1}

3.609
answers 24 and 27
3/1/79
63

line 3 of answer 24: replace by lines 8 and 9 of answer 27

lines 8 and 9 of answer 27 should be:

$\alpha \neq \beta$; $g(x) = x^\beta(\ln x + C)$ for $\alpha = \beta$. We have $p_t(-t-2) = 0$; so the general solution to our differential equation is

3.624	line -5	5/1/79	64
	$g_{M,N}^{n+1}(z) \rightsquigarrow g_{M,N}^{(n+1)}(z)$		
3.675	improvement to 1979 change #312	1/31/79	65
	produces $\rightsquigarrow$ results in (twice)		
3.683	answer 14 line 4	1/31/79	66
	TAG $\rightsquigarrow$ TAG		
3.688	new answer 10	3/1/79	67
	10. A simple proof has been found by Frank M. Liang [<i>Discrete Math.</i> , to appear].		
3.689	new paragraph inserted at beginning of answer 14	2/23/79	68
	14. According to the stated conventions, the notation “ $X \Leftarrow \text{AVAIL}$ ” of 2.2.3–6 now stands for the following operations: “Set $X \leftarrow \text{AVAIL}$; then set $X \leftarrow \text{LINK}(X)$ zero or more times until either $X = 0$ (an OVERFLOW error) or $\text{TAG}(X) = 0$; finally set $\text{AVAIL} \leftarrow \text{LINK}(X)$.”		
3.689	new paragraph appended at end of answer 14	2/23/79	69
	Another way to place a hash table “on top of” a large linked memory, using coalescing lists instead of separate chaining, has been suggested by J. S. Vitter (to be published).		
3.710R		10/18/79	70
	Anuyogadvarā $\rightsquigarrow$ Anuyogadvāra		
3.712L	delete 1979 change #334	3/1/79	71
	(Fan Chung no longer mentioned on page 688)		
3.716R		3/1/79	72
	Liang, Frank Mark, 688.		
3.719L	Vaughan Pratt entry	1/24/79	73
	add p. 450		

3.721L update to 1979 change #384

3/1/79 74

Sprugnoli, Renzo, 507.

9.0 changes for the book Mariages Stables

6/1/79 75

p. 91 lines -4 thru -6 should be:

$$Ac, Ff \Rightarrow Ee, Dd, Ba, Cb;$$

$$Ac, Fd \Rightarrow Ef, De, Ba, Cb;$$

$$Ac, Fe \Rightarrow Ba, Df, Cb, Ed.$$

p. 92: here the two 3×3 diamonds should not have a common point, they should be attached by a vertical line, and the text should be modified accordingly to state that it is the direct sum of a 2-element chain with two 3×3 diamonds